Sheng Zhao

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*My research interests focus on developing efficient, controllable and truthworthy **computing systems** facilitating interaction between **humans** and **mobile devices**, leveraging **AI** and **ML** technologies.*

Education

Tsinghua University, Beijing, China 2021.09 – Present
B.Sc. in **Mathematics and Physics** & B.Eng. in **Electrical Engineering** Dual degree

- Junior, GPA: 3.81/4.
- A+: Optics, Quantum Mechanics; A: Electromagnetism, Thermodynamics, Calculus, Mathematical Physics Equations, etc.

Publications & Manuscripts

- **Sheng Zhao**, JunRay Zhu, Xueyang Wang, Hongyi Li, Fang Yi, Hewu Li†, Shuning Zhang† and Xin Yi†. CoordAuth: A Two-factor Authentication Method in Virtual Reality Leveraging Head-Eye Coordination. *Ubicomp'24, under rewiew.* PDF.
- Xueyang Wang*, **Sheng Zhao***, Yihe Wang, Ziyu Han, Xin Tong† and Xin Yi†. EmoBlend: Revealing the Impact of Facial Blendshape Intensity on Emotional Perception in Virtual Reality. *TVCG, under rewiew*.

  *Equal contribution. †Corresponding author.

Research Experiences

EmoBlend: Revealing the Impact of Facial Blendshape Intensity on Emotional Perception in VR 2023.04 – 2024.01
Research Assistant, Advisor: *Xin Yi, Yuanchun Shi* Tsinghua University

- Implementing VR face2face chat using TCP-based socket and Oculus-Movement sdk in Unity.
- Designing both qualitative and quantitative research methods based on interviews and questionnaires, conducting user studies.
- Proposing an improved camera2blendshape method to mitigate the uncanny valley effect and enhance the emotional experience.
- Submitted to *TVCG*. Details.

CoordAuth: A Two-factor Authentication Method in VR Leveraging Head-Eye Coordination 2023.10 – 2024.01
Research Assistant, Advisor: *Xin Yi* Tsinghua University

- Project leader, proposing the current two-factor authentication system based on unlock pattern and behavior biometrics.
- Using Unity to build the unlock UI leveraging gaze interation in VR.
- Proposing a voting mechanism using two types of classifiers implemented by random forest to achieve behavioral biometric authentication factor.
- Conducting user studies, validating the robustness against password collision, contextual factors, and shoulder surfing attacks.
- Submitted to *Ubicomp'24*. PDF. Details.

AvatarMVFusion: Text-Guided Multi-View Diffusion for Avatar Generation 2024.01 – Present
Research Assistant, Advisor: *Bowen Zhou* Tsinghua University

- Fine-tuning diffusion model by dreambooth for avatar generation with high quality. Details.
- Testing the performance of Zero123 (SOTA) and DreamFusion, figuring out the importance of prompt-driven multi-view generation for the avatar leveraging diffusion.
- Proposing a novel text-guided denoising Unet framework and conducting experiments to evaluate its performance.

EVDrive: Utilizing EventCamera to Augment Auto Driving Perception 2024.01 – Present
Remote Research Intern, Advisor: *Yukang Yan* University of Rochester

- Reproducing EV-Eye (NIPS'23) using Unet and polynomial fitting of gaze directions.
- Developing an auto-driving simulation environment using Unity 3D.
- Designing a ML architecture integrated into the EV-Eye framework to achieve perception of the driver's state.

Skills

- Program languages: C/C++/C#, Python, Java, MATLAB and Verilog.
- Tools and Frameworks: Git, LaTeX, Unity, PyTorch, Scikit-learn, Markdown and MSP430.
- Techniques: VR development, Quantitative and Qualitative research.

Competition Awards

- *First Prize*, 36th China Physics Olympiad (CPHO), 0.9% 2020
- *First Prize*, The 12th Tsinghua University Digital System Design Competition, 1.7% 2022

  Designed a small car driven by MSP430 with various sensors. Details.
- *Second Prize*, Tsinghua University Software Design Competition 2023

  Developed a Smart Home app using Java. Code.
- *First Prize*, The 3rd Tsinghua Craftsman Competition, 1%, presented a speech 2023